Solutions to MP352 Problem Set 2

(1)

P.1

(a) We can see how fast our two mecha pilots are going simply by dividing the times by one another: recall that if an observer in frame S sees a clock in frame S' moving at a speed v, she sees the clock tick more slowly because of the time dilation effect. In other words, if Δt is a time interval in S and $\Delta t'$ one in S', they are related by $\Delta t' = \frac{\Delta t}{\gamma(v)}$. In this case, S' is Noriko and Kazumi's frame, where $\Delta t' = 2$ days. S is the rest frame of Earth, where $\Delta t = 12000\text{ years} = 12000 \times 365$ days $= 4.38 \times 10^6$ days. Therefore

$$\gamma(v) = \frac{\Delta t}{\Delta t'} = 2.19 \times 10^6$$

which is huge. This indicates already that v is extremely close to the speed of light. Thus, it's probably better to compute $1 - v/c$ rather than v/c itself. This can be done exactly, but let's do it a different way that will take less work: $\gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}$, so $1 - v^2/c^2 = \frac{1}{\gamma^2(v)}$, or $\frac{v}{c} = \sqrt{1 - 1/\gamma^2(v)}$. Now, γ^2 is huge, so $1/\gamma^2$ is extremely small ($\sim 10^{-12}$). Thus, we can use the Taylor series expansion $\sqrt{1-x} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2 + \dots$ Since $\frac{1}{\gamma^4}$ is 10^{-12} times smaller than $1/\gamma^2$, we can take $\sqrt{1 - 1/\gamma^2} \approx 1 - \frac{1}{2\gamma^2(v)}$ to get a very good approximation of v/c. Using this, we see

$$1 - v/c \approx \frac{1}{2\gamma^2(v)} = \boxed{1.04 \times 10^{-13}}$$

In other words, Noriko and Kazumi are travelling at only 31.3 micrometers per second less than the speed of light. Now THAT's fast!

(b) But we not only have time dilation, we also have length contraction. It takes the women only two days because, to them, the distance to Earth is much less than the distance that someone on Earth sees to them. More precisely, if L is the distance an Earth-bound observer sees, $L' = L/\gamma(v)$ is the distance Noriko and Kazumi see to Earth.

Now, for all intents and purposes, each sees the other as travelling at nearly the speed of light. Thus, if Earth sees it taking 12000 years, then this means $\boxed{L = 12000\text{ ly}}$ is how far away Noriko & Kazumi are. But if they see it as taking only two days for the Earth – also travelling at near c – to get to them, then they see themselves travelling a distance $2\text{ days} \times c$, i.e. two light-days, or $\frac{2}{365}\,ly = \boxed{5.48\times10^{-3}\,ly}$. (In case you're interested, this is about 5.2×10^{10} km, or 347 astronomical units. This is about three times the distance Voyager 1 has travelled so far, or about one third the distance to 90377 Sedna at aphelion.)

P.2

The reason for this problem lies in a calculation that I did in lecture, that two consecutive boosts in the x-direction, one of speed v and one of u', give the same result as a single boost of speed $u = \frac{u'+v}{1+\frac{u'v}{c^2}}$.
This proof works only if

$$\gamma(u) = \gamma(u')\gamma(v)\left(1+\frac{u'v}{c^2}\right),$$

so I figured I'd ask you to prove that here.
We first compute $\gamma(u)$; it is

$$\gamma(u) = \frac{1}{\sqrt{1-u^2/c^2}} = \frac{1}{\sqrt{1-\frac{1}{c^2}\left(\frac{u'+v}{1+\frac{u'v}{c^2}}\right)^2}} = \frac{1}{\sqrt{1-\frac{(u'+v)^2c^2}{(c^2+u'v)^2}}}$$

$$= \sqrt{\frac{(c^2+u'v)^2}{(c^2+u'v)^2-(u'+v)^2c^2}} = \frac{c^2+u'v}{\sqrt{[c^4+2c^2u'v+(u')^2v^2]-[(u')^2+2u'v+v^2]c^2}}$$

Now, the quantity under the square root is

$$\begin{aligned}(c^2+u'v)^2-(u'+v)^2c^2 &= [c^4+2c^2u'v+(u')^2v^2]-[(u')^2+2u'v+v^2]c^2 \\ &= c^4+(u')^2v^2-(u')^2c^2-v^2c^2 \\ &= [c^2-(u')^2][c^2-v^2]\end{aligned}$$

as you can easily check. This is $c^4\left(1-\frac{(u')^2}{c^2}\right)\left(1-\frac{v^2}{c^2}\right) = \frac{c^4}{\gamma^2(u')\gamma^2(v)}$.
Thus,

$$\gamma(u) = \frac{c^2+u'v}{\sqrt{c^4/\gamma^2(u')\gamma^2(v)}} = \frac{\gamma(u')\gamma(v)}{c^2}\cdot(c^2+u'v) = \boxed{\gamma(u')\gamma(v)\left(1+\frac{u'v}{c^2}\right)}$$

and so we're done.

P.3

This is kind of similar to a problem in PS1, the one where we checked that the general forms for t' and $\vec r'$ reduced to the usual Lorentz transformations for a boost in the positive x-direction. We do the same here by taking $\vec V = V\hat e_x$. When this is the case, we see that

③

$\vec{u}' \cdot \vec{v} = u_x' v$, so $1 + \frac{\vec{u}' \cdot \vec{v}}{c^2} = 1 + \frac{u_x' v}{c^2}$ and $\left(\frac{\vec{u}' \cdot \vec{v}}{v^2}\right) \vec{v} = u_x' \hat{e}_x$.

Then,

$$\vec{u} = \frac{1}{\gamma(v)\left(1 + \frac{\vec{u}' \cdot \vec{v}}{c^2}\right)} \left[\vec{u}' + \gamma(v) \vec{v} + (\gamma(v) - 1) \frac{(\vec{u}' \cdot \vec{v})}{v^2} \vec{v}\right]$$

$$= \frac{1}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)} \left[u_x' \hat{e}_x + u_y' \hat{e}_y + u_z' \hat{e}_z + \gamma(v) v \hat{e}_x + (\gamma(v) - 1) u_x' \hat{e}_x\right]$$

$$= \frac{1}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)} \left(u_x' + \gamma(v) v + (\gamma(v) - 1) u_x'\right) \hat{e}_x$$

$$+ \frac{1}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)} \left(u_y'\right) \hat{e}_y + \frac{1}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)} \left(u_z'\right) \hat{e}_z$$

so

$$\boxed{u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}, \quad u_y = \frac{u_y'}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)}, \quad u_z = \frac{u_z'}{\gamma(v)\left(1 + \frac{u_x' v}{c^2}\right)}},$$

exactly what we found in lecture.